

CSA1006-SOFTWARE ENGINEERING

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CO1 AT-1

CASE STUDY

AI-Based Smart Hospital Appointment and Patient Management System

1. Understanding the Problem Statement

Problem Explanation

A multi-specialty hospital serves thousands of patients every day. Patients experience long waiting times, manual appointment booking, misplaced medical records, delayed doctor availability updates, and poor communication between departments. The hospital wants to develop an AI-Based Smart Hospital Appointment and Patient Management System that enables online appointment booking, AI-based appointment scheduling, digital medical records, real-time notifications, and automated report generation.

Objectives

- Develop an online appointment booking system.
- Automate appointment scheduling using AI.
- Manage patient medical records digitally.
- Reduce patient waiting time.
- Generate hospital performance reports.

Scope

- Patient registration
- Doctor management
- Online appointment booking

- AI appointment scheduling
- Electronic Medical Records (EMR)
- Billing management
- Notification system
- Admin dashboard
- Report generation

2. Stakeholder Analysis

Stakeholder	Responsibilities
Patients	Register, book appointments, view reports
Doctors	Manage appointments and patient records
Receptionists	Verify appointments and billing
Laboratory Staff	Upload test reports
Hospital Administrator	Manage users and hospital operations
Hospital Management	Monitor hospital performance

Challenges

- Long waiting time
- Manual appointment booking
- Duplicate patient records
- Poor communication
- Delayed report generation
- Difficulty tracking appointments

3. Current System Analysis

Existing Process

- Patients visit hospital physically.
- Receptionist books appointments manually.
- Patient records stored on paper.
- Reports generated manually.
- Doctors update records manually.

Strengths

- Easy to understand
- No internet dependency
- Low initial cost

Limitations

- Time-consuming
- Human errors
- Paper record management
- Duplicate patient files
- Long patient waiting time
- Difficult report generation

Problems

- Appointment conflicts
- Lost patient records
- Delayed treatment
- Poor resource allocation
- Inefficient communication

4. Proposed Software Solution

Proposed System

Develop a Web-Based AI Smart Hospital Management System that automates appointments, patient records, billing, notifications, and report generation.

Functional Modules

Module 1: User Management

- Patient Registration
- Doctor Login
- Reception Login
- Admin Login

Module 2: Appointment Management

- Book Appointment
- Cancel Appointment
- Reschedule Appointment
- View Doctor Availability

Module 3: AI Scheduling Module

- Automatic appointment allocation
- Priority for emergency patients
- Waiting time prediction

Module 4: Medical Records

- Digital patient records

- Prescription management
- Laboratory reports
- Medical history

Module 5: Billing Module

- Online payment
- Invoice generation
- Insurance details

Module 6: Dashboard

- Daily appointments
- Patient statistics
- Doctor schedules
- Revenue reports

Module 7: Notification System

- SMS reminders
- Email notifications
- Appointment alerts

Module 8: Report Generation

- Daily Reports
- Monthly Reports
- Doctor Performance Reports
- Patient Statistics

Functional Requirements

- User Registration
- Secure Login
- Appointment Booking
- Appointment Tracking
- AI Scheduling
- Medical Record Management
- Billing
- Notifications
- Report Generation

Non-Functional Requirements

- Fast response time
- High availability
- Secure authentication
- Reliable database
- User-friendly interface
- Scalability
- Data backup
- Privacy protection

Software Engineering Principles Used

Modularity

Separate modules for appointments, billing, reports, and notifications.

Scalability

Supports increasing patients and doctors.

Maintainability

Modules can be updated independently.

Reliability

Automatic backup and data validation.

Usability

Simple interface for patients and hospital staff.

Security

Password encryption, role-based access, and secure medical records.

5. SDLC Justification

Selected Model

Agile SDLC

Why Agile?

- Frequent requirement changes
- Continuous feedback from hospital staff
- Easy feature addition
- Faster software delivery
- Continuous testing

Agile Process

Requirement Gathering



Planning



System Design



Development



Testing



Deployment



Maintenance

6. Expected Outcomes

Benefits

- Faster appointment booking
- Reduced patient waiting time
- Better doctor scheduling
- Digital medical records
- Improved hospital efficiency
- Better communication
- Increased patient satisfaction

Addressed Challenges

- Eliminates paper records
- Reduces appointment conflicts
- Provides real-time updates
- Improves communication
- Enhances report generation
- Better resource utilization

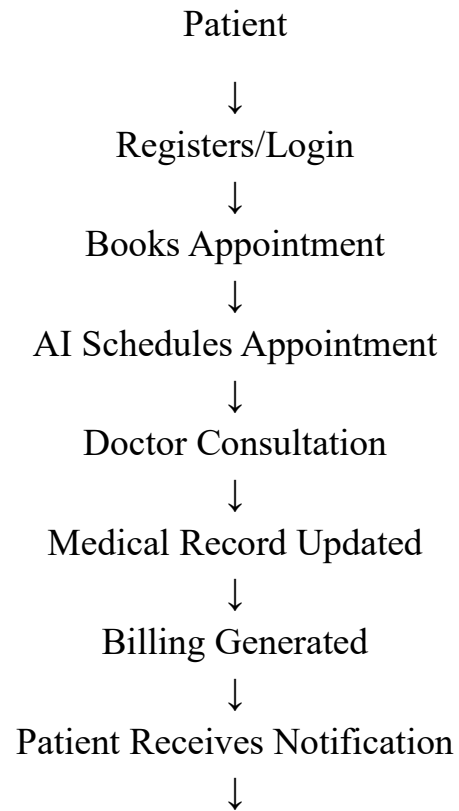
Risks

- Internet dependency
- Initial implementation cost
- User training required
- Data privacy concerns
- AI scheduling prediction errors

Future Enhancements

- Mobile application
- AI chatbot for patient support
- Telemedicine integration
- Voice-based appointment booking
- Wearable health device integration
- Predictive disease analysis

Output



Reports Generated



Conclusion

The AI-Based Smart Hospital Appointment and Patient Management System provides an efficient, secure, and scalable solution for managing hospital operations. By applying Agile SDLC and software engineering principles such as modularity, maintainability, scalability, usability, reliability, and security, the system improves appointment scheduling, reduces waiting time, enhances communication, and increases patient satisfaction. Future enhancements including AI chatbots, telemedicine, and predictive healthcare will make the system even more intelligent and efficient.